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CHAPTER 1: INTRODUCTION

1.1 Project Background

Nowadays, traditional badminton court booking methods can lead to issues such as double booking and poor data management. Hence, online systems for booking court play an important role in improving efficiency and user convenience.

The Court Badminton Reservation System is developed as a web-based application. This system allows users to book badminton courts easily by checking available courts and time slots in real time. It also enables admin to manage reservations and updating data in real time.

1.2 Problem Statements

Traditional booking systems face several issues:

- High risk of double booking.
- Inefficient data management.
- Difficulty in checking real-time court availability.
- Difficulty to check their reservations.

Hence, a web-based reservation system is needed to improve booking efficiency and accuracy.

1.3 Objectives

- To make booking badminton courts easier and more convenient by providing available courts and time slots.
- To develop a web-based system that enables users to view their reservations in real time.
- To allow admin to add, delete and manage court reservations efficiently.
- To manage booking and payment data accurately and systematically.

1.4 Project Scope and Target User

Project Scope:

- User registration and login system.
- Viewing available courts and time slots.
- Making reservations.
- Updating and cancelling reservations.
- Payment confirmation feature.
- Admin management of reservation data.

Target User:

- Public User: Individuals who want to book badminton courts easily and conveniently.
- Admin User: System administrators responsible for managing reservations and maintaining system data.

CHAPTER 2: SYSTEM REQUIREMENTS

2.1 Functional Requirements

User Module:

- Register and log in to the system.
- View available courts and time slots.
- Make court reservations.
- View their reservations details.
- Update or cancel reservations.
- Confirm payment.

Admin Module:

- View all reservation cancelled.
- Update reservations details.
- Delete reservation records.
- Manage booking and payment confirmation.

2.2 Non-Functional Requirements

- Performance: System should respond quickly when users access or update data.
- Security: User information such as username and password must be protected.
- Usability: The interface should be simple and user-friendly.
- Reliability: System should operate without errors and downtime.
- Maintainability: System should be easy to maintain and update.

2.3 Hardware and Software Requirements

Hardware:

- Laptop
- Internet connection

Software:

- XAMPP
- Visual Studio
- Web browser
- Programming languages: HTML, PHP, CSS
- Database: MySQL

2.4 Data Flows

1. Users log into system.
2. User select court, date and time slot.
3. System check availability.
4. Reservation is stored in the database.
5. User can update, cancel or confirm payment.
6. Admin can view and manage such as add, delete, update all reservation data.

All data is stored in database ensuring real time updates.

CHAPTER 3: CONCLUSION

3.1 Achievements

The web-based badminton court reservation system successfully developed with the following features:

- Easy booking system with real-time availability.
- Reservation management like add, update, cancel and delete.
- Admin management system.

The system can reduces booking conflicts compared to traditional methods.

3.2 Project Evaluation

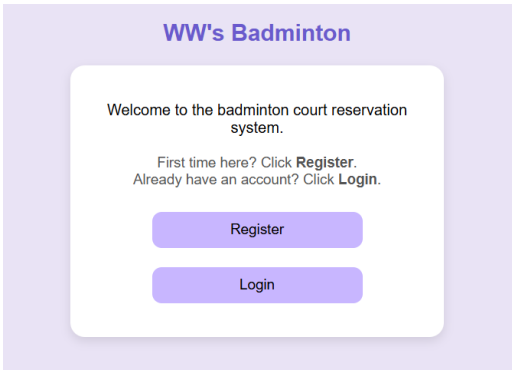
The system meets its objectives by providing a functional reservation system. It simplifies the booking process and improves data management.

Suggestions for improvement:

- Improve security features like password hashing.
- Improve user interface.
- Real-time slot availability.
- Confirmation message through email outside the system.

4.0 APPENDIX

4.1 Homepage



4.2 Registration Page

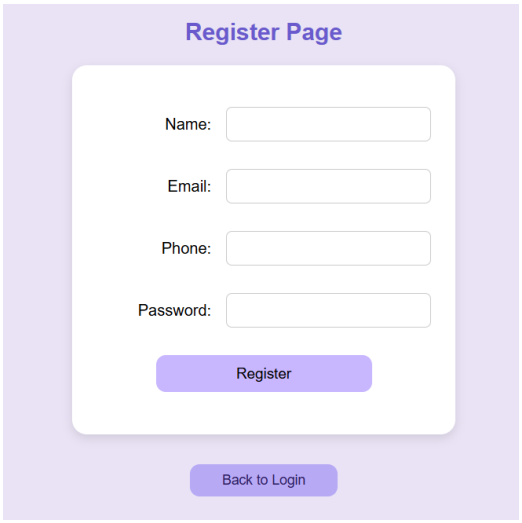


Figure 1: Registration user interface

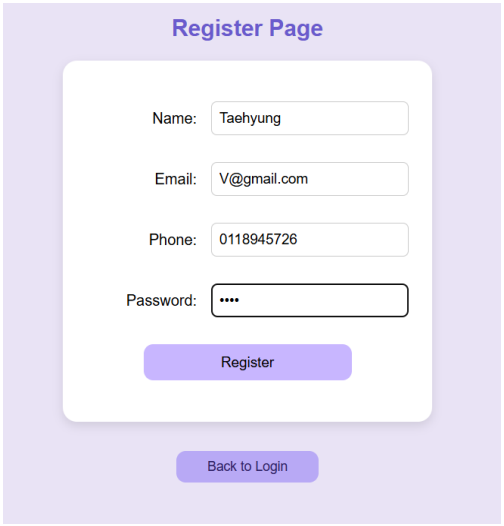


Figure 2: User insert information

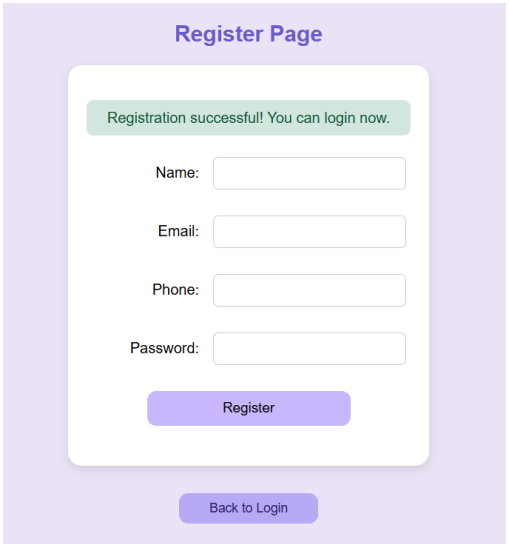


Figure 3: Registration successful confirmation

4.3 User Page

4.3.1 User Login

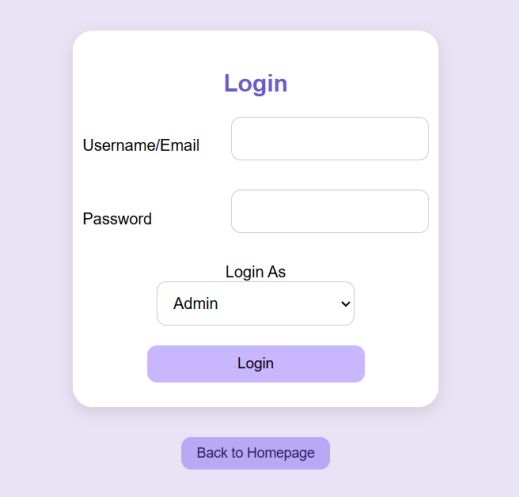


Figure 4: Login Interface

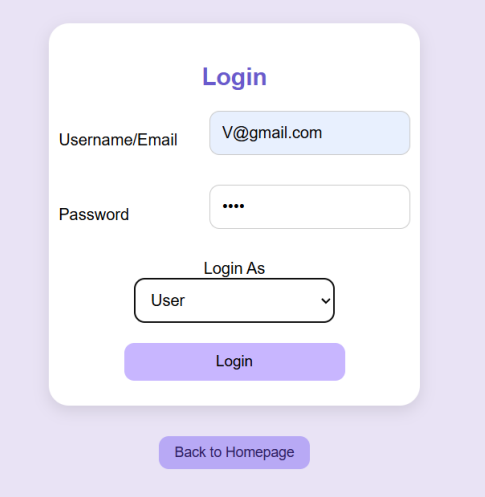


Figure 5: User login

4.3.2 User Menu

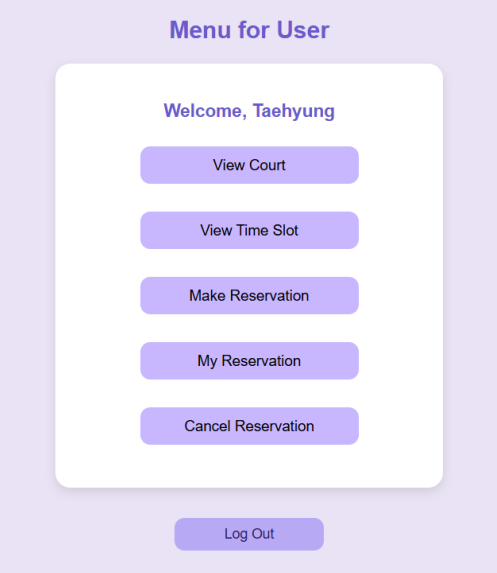


Figure 6: User Menu

4.3.3 View Court

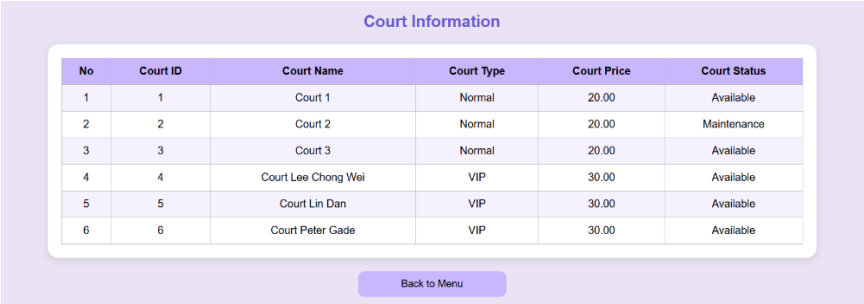


Figure 7: Court Information

4.3.4 View Time Slot

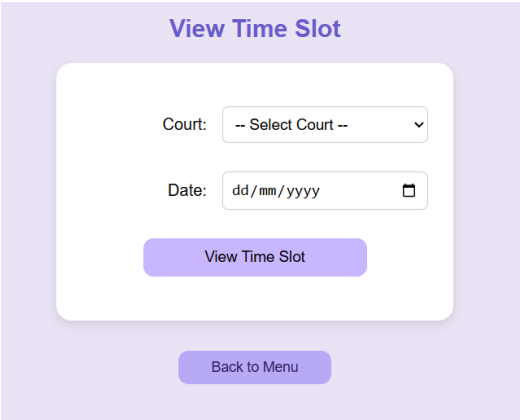


Figure 8: Time Slot Interface

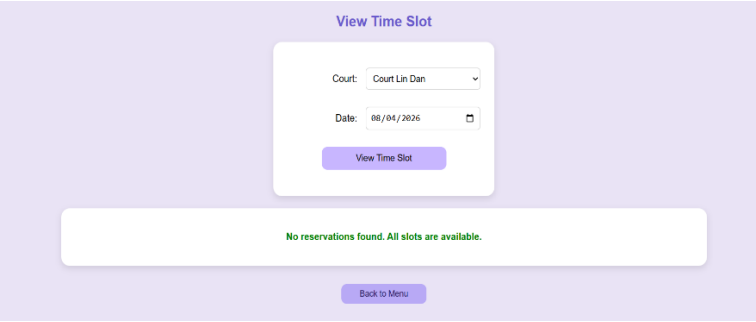


Figure 9: No booking on the court and date chosen

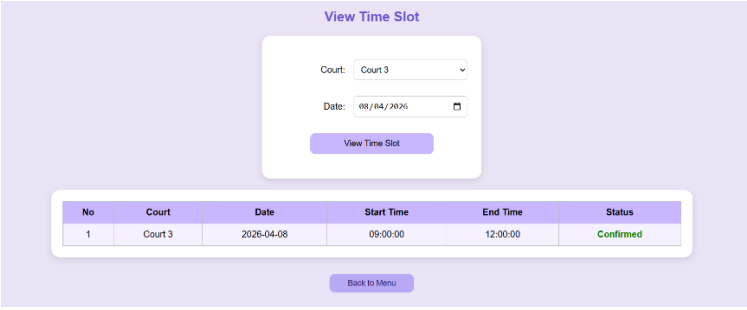


Figure 10: Have booking on the court and date chosen

4.3.5 Make Reservation

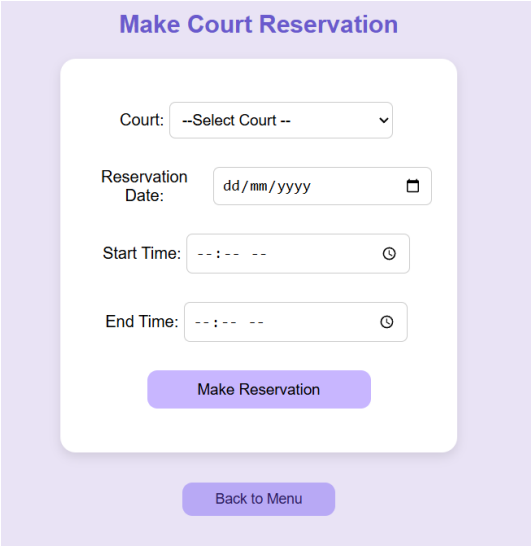


Figure 11: Court Reservation Interface

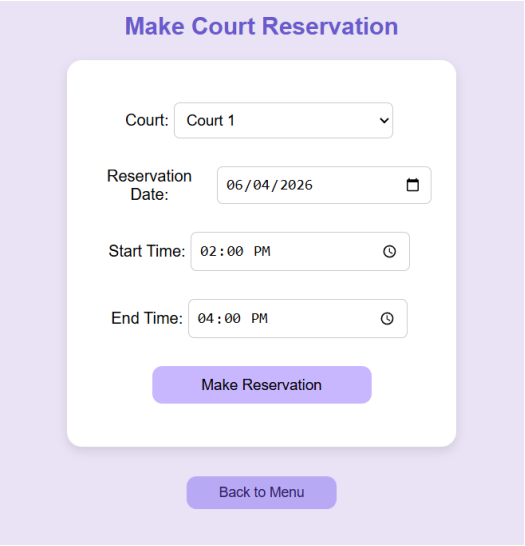


Figure 12: User choose court and date

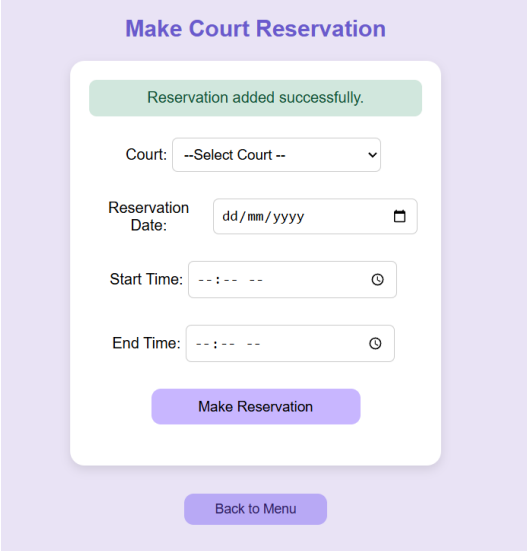


Figure 13: Confirmation message

4.3.6 View My Reservation

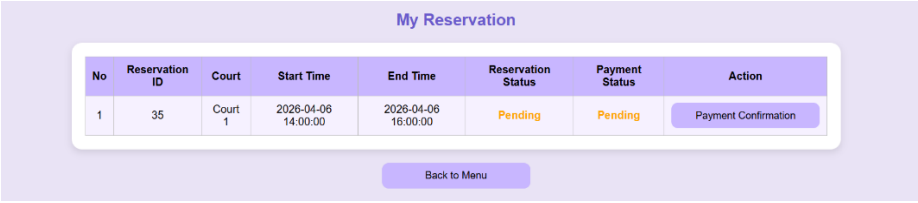


Figure 14: Reservation details

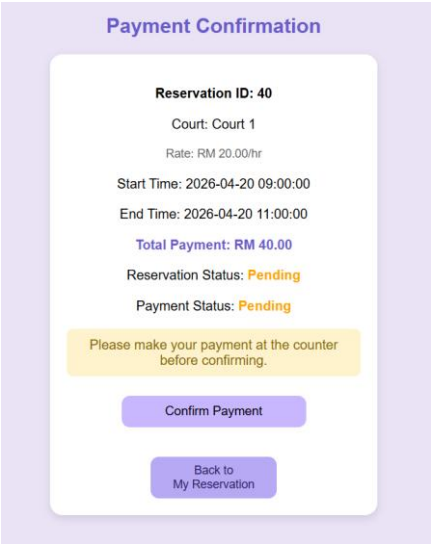


Figure 15: Confirmation for payment and reservation



Figure 16: Reservation and payment confirm

4.3.7 Cancel Reservation

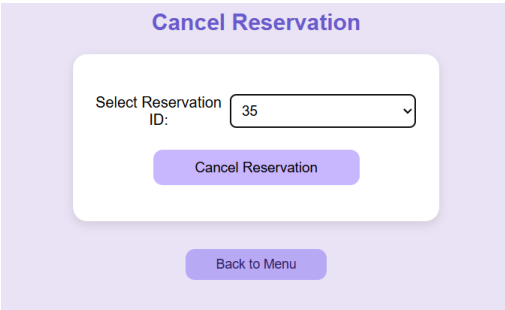


Figure 17: Choose reservation ID



Figure 18: Ask for confirmation

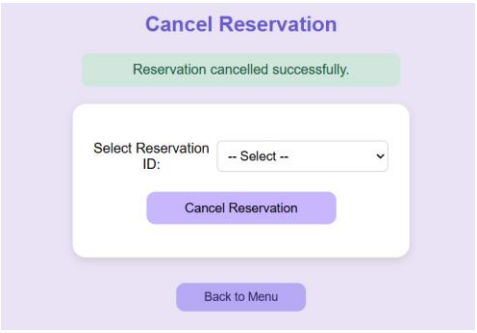


Figure 19: Confirmation and cancelled



4.4 Admin Page

4.4.1 Admin Login

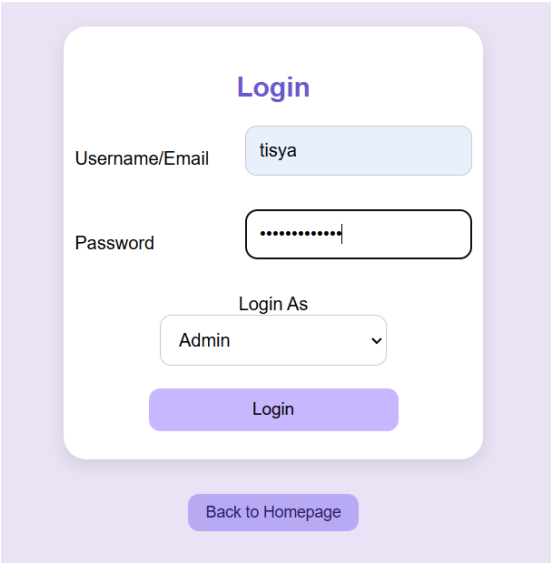


Figure 20: Admin login

4.4.2 Admin Menu

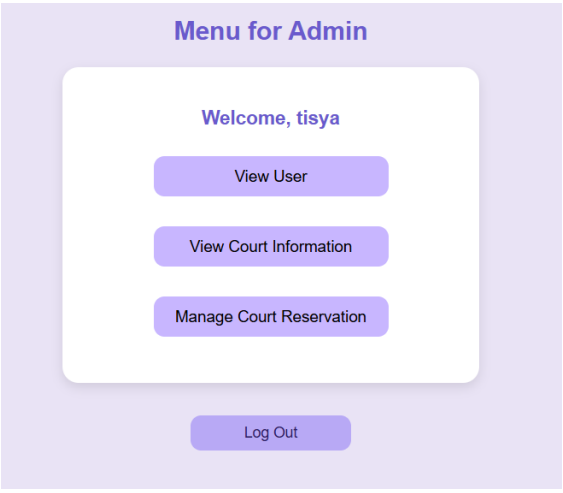


Figure 21: Admin Menu

4.4.3 View User



Figure 22: List of registered users

4.4.4 Court Information

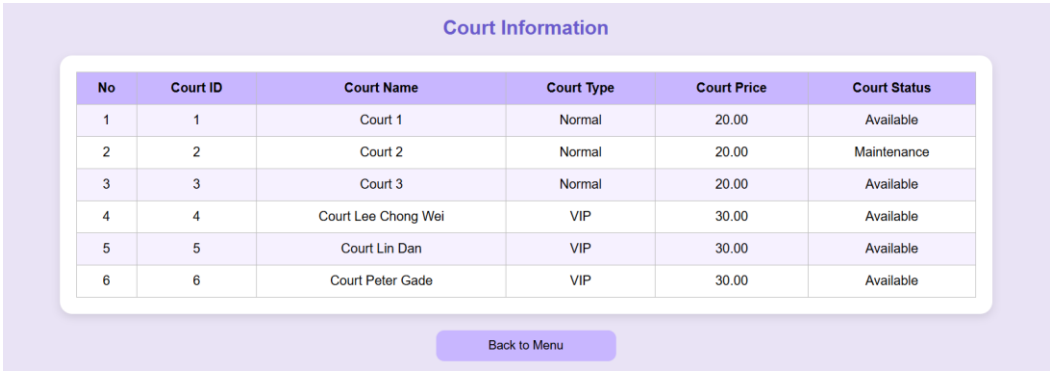


Figure 23: Court Information

4.4.5 Manage Court



Figure 24: Admin role

4.4.6 Add Reservation

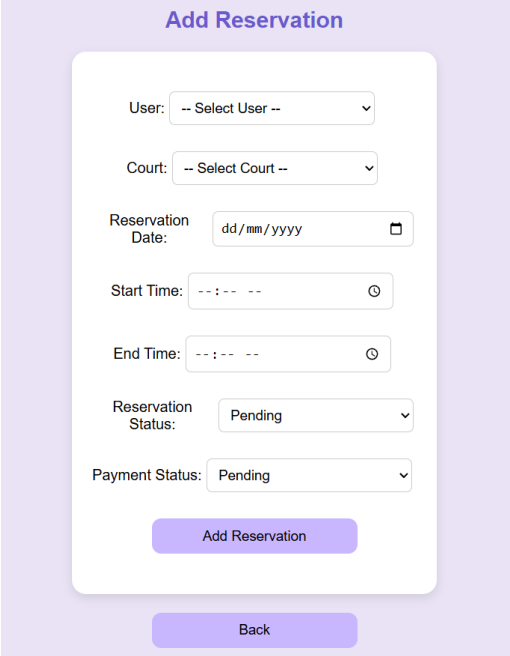


Figure 25: Add reservation interface

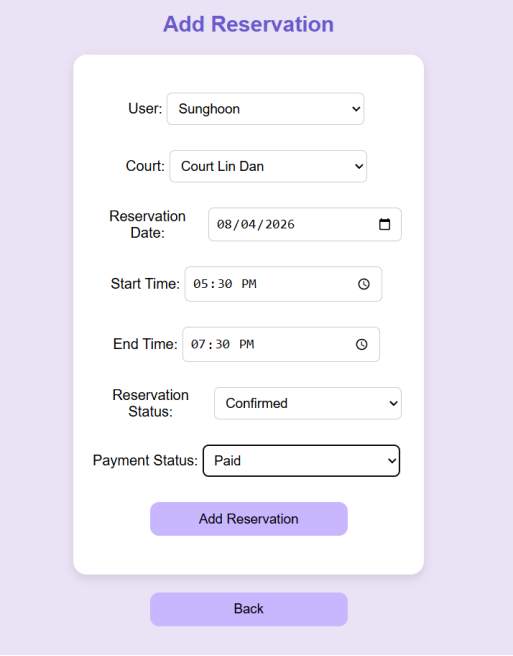


Figure 26: Insert information

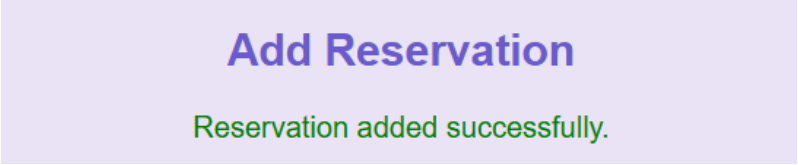


Figure 27: Confirmation message

4.4.7 View Reservation



Figure 28: List of reservation

4.4.8 Update Reservation

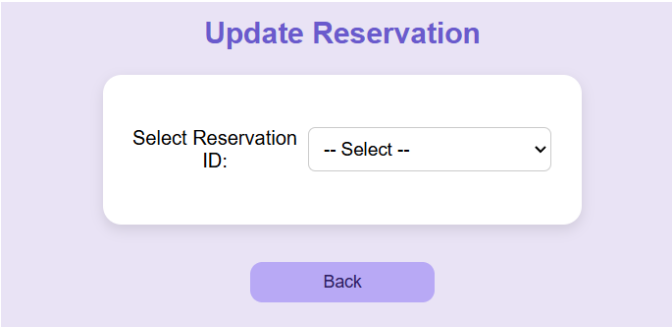


Figure 29: Update interface

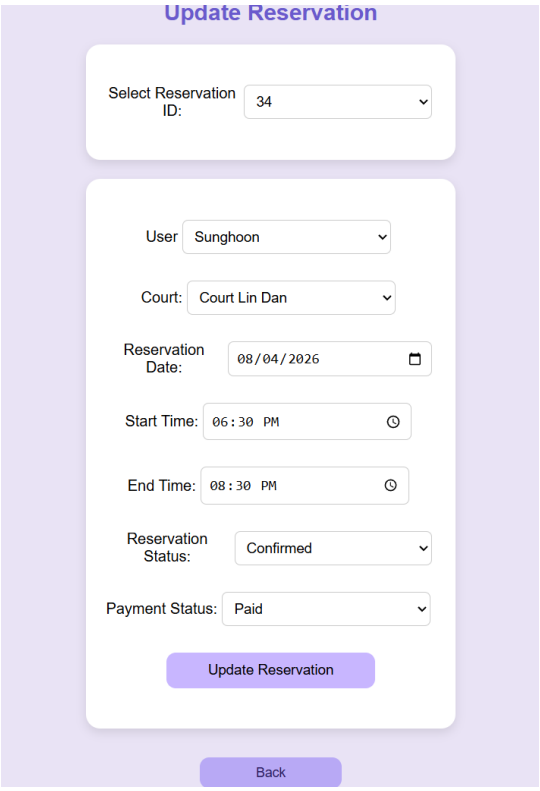


Figure 30: Choose info to update

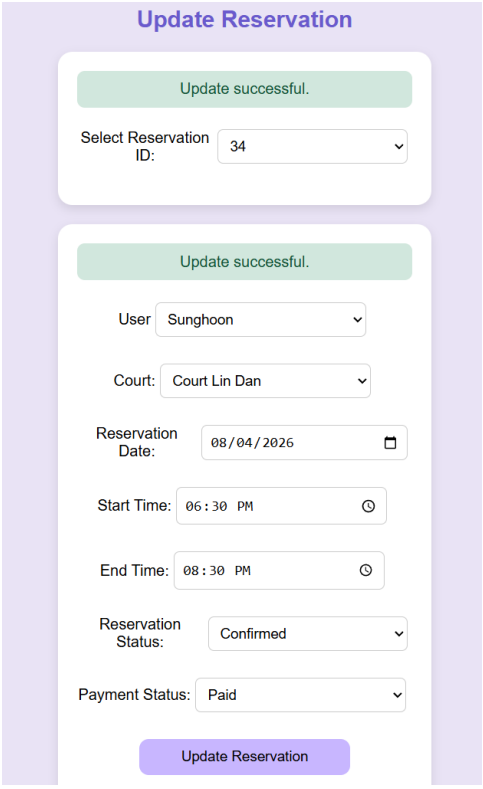


Figure 31: Confirmation message

4.4.9 Delete Reservation

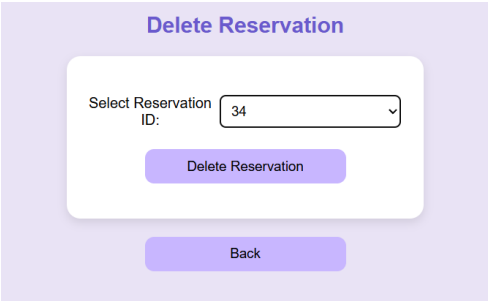


Figure 32: Delete interface

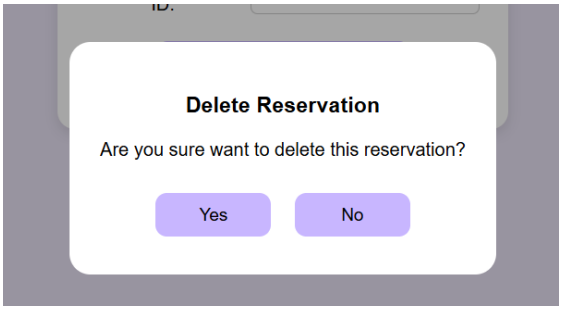


Figure 33: Ask for confirmation

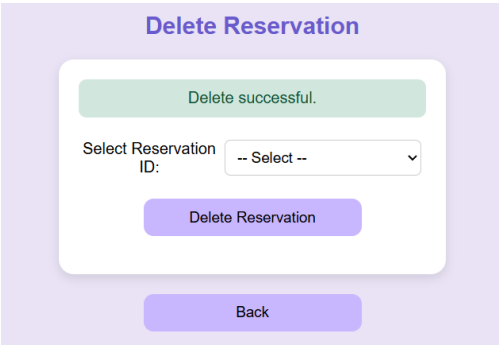


Figure 34: Confirmation message

4.4.10 Search Reservation

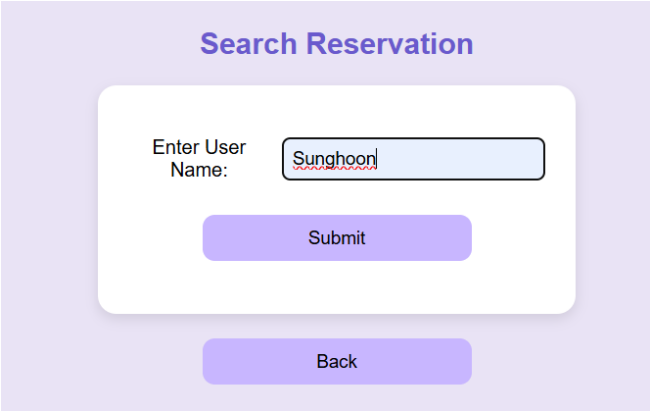


Figure 35: Search interface

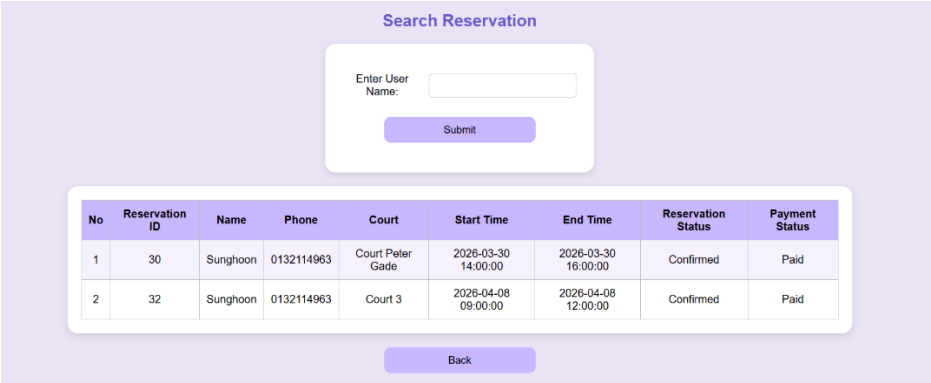


Figure 36: List of reservation under user name

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